

Project Proposal

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Title: Forecasting Football: A Comparative Analysis of Classification Models for Premier League Match Predictions

Problem Statement

Football (or soccer) is arguably the world's most watched sport, and the English Premier League (EPL) is the most watched league. This project will aim to develop and evaluate a series of machine learning models in order to classify the outcome of a match in the EPL into one of three categories, i.e. Home Win, Draw, or Away Win. The primary goals would involve:

1. Building and rigorously comparing the predictive accuracy of multiple classification models, from linear models to ensemble methods.
2. Identifying the most influential factors in determining the outcome of a match. By contrasting the feature importance derived from the different model paradigms, we can get a multi-angled, multi-faceted look at what statistically drives results.

Data Source

<https://football-data.co.uk/englandm.php> - Provides historical match datasets for multiple seasons, dating all the way back to 1993. This project will utilize data starting from the 2014-2015 season and stopping at the 2024-2025 season, the last 10 years. The 2025-2026 is still underway, so it would be considered an incomplete dataset.

Methodology

This project will consist of four main steps. The first step is data preprocessing and cleaning, which involves the standardization of the numeric features in order to zero their mean and unit variance. This step is crucial for SVM and Logistic Regression. To detect outliers, the project will be checked via the geometric and the statistical method. The geometric method uses one-class SVM to find a boundary that encloses on a majority of the data points. The statistical method, on the other hand, uses Kernel Density Estimation (KDE) to model the probability density of the data. Points in low density regions will be marked as outliers potentially.

The second step is feature engineering and dimensionality reduction. In order to get the team form and momentum data, time-dependent features would need to be produced, including performance differentials and rolling averages. These newer features may contain invalid data points or complex non-linear data, and to counteract this, firstly, PCA will be applied to find the

linear projections that maximize variance. Secondly, to confirm whether or not the data is on a complex non-linear projection, ISOMAP will be implemented. Isometric Feature Mapping estimates the intrinsic geometry of the data by using graph based geodesic distances. Based on which technique is the most optimal, the low dimension components that come out of it will be utilized as another feature set for the classifier models to compare and contrast for any performance differences.

The third step is model selection and training. The models I suggest in this proposal are Multiclass Regression with Lasso Regularization, Support Vector Machine (SVM), Random Forest Classifier, Gradient Boosting (XGBoost). The dataset would be split chronologically and using cross-validation on the training dataset, hyperparameter tuning can be done. Log-loss will be the primary metric for optimizing the hyperparameters. The final models will be tested on the 2024-2025 season, i.e. the hold-out dataset.

Final Results and Evaluation

Lastly, after the models have been evaluated and checked on the hold-out data through multiple metrics, and they would be the overall accuracy, confusion matrix, precision, recall and/or F1 score, and Log-loss. To ensure a robust analysis, finding which factors are most predictive of the match outcome is paramount. To find these, linear interpretability (through Lasso Regression) and ensemble interpretability (through random forest and XGBoost), along with Mutual Information. By comparing and contrasting the results from these methods, a strong conclusion can be reached on the statistics that decide Premier League games.